



Rossmoyne Senior High School

Semester One Examination, 2022

Question/Answer booklet

MATHEMATICS METHODS UNIT 1

Section One: Calculator-free

If required by your examination administrator, please
place your student identification label in this box

WA student number: In figures

--	--	--	--	--	--	--	--

In words

Your name

--

Circle your Teacher's Name:

Mr Barrymore

Mrs Bestall

Mr Buckland

Ms Chua

Mrs Fraser-Jones

Mr Koulianos

Mr Mohammadi

Ms Leonard

Mr Luzuk

Time allowed for this section

Reading time before commencing work: five minutes
Working time: fifty minutes

Number of additional
answer booklets used
(if applicable):

--

Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet
Formula sheet

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener,
correction fluid/tape, eraser, ruler, highlighters

Special items: nil

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	7	7	50	54	35
Section Two: Calculator-assumed	12	12	100	96	65
Total					100

Instructions to candidates

1. The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.
2. Write your answers in this Question/Answer booklet preferably using a blue/black pen. Do not use erasable or gel pens.
3. You must be careful to confine your answers to the specific question asked and to follow any instructions that are specific to a particular question.
4. Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.
5. It is recommended that you do not use pencil, except in diagrams.
6. Supplementary pages for planning/continuing your answers to questions are provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.
7. The Formula sheet is not to be handed in with your Question/Answer booklet.

Section One: Calculator-free**35% (54 Marks)**

This section has **seven** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 50 minutes.

Question 1**(6 marks)**

Solve each of the following equations.

(a) $\frac{x}{2} + \frac{2x}{3} = \frac{x+1}{2}$. (2 marks)

(b) $3x^3 = 12x^2$. (2 marks)

(c) $(x+5)^2 - 49 = 0$. (2 marks)

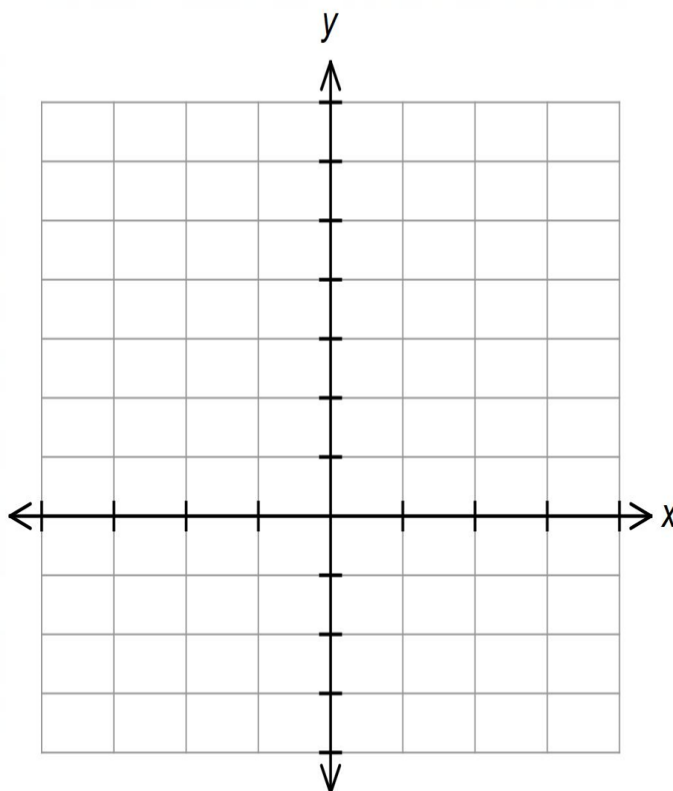
Question 2

(6 marks)

(a) Solve the equation $6 - x^2 = x$.

(2 marks)

(b) Sketch the graphs of $y = 6 - x^2$ and $y = x$ on the axis below, showing the coordinates of all axes intercepts of the parabola and any points of intersection of the graphs. (4 marks)



Question 3

(8 marks)

(a) Expand $(x + 4)(x - 1)(x - 4)$.

(2 marks)

(b) Let $f(x) = x^3 - x^2 - 14x + 24$.(i) Calculate $f(-2) - f(2)$.

(2 marks)

(ii) Solve $f(x) = 0$.

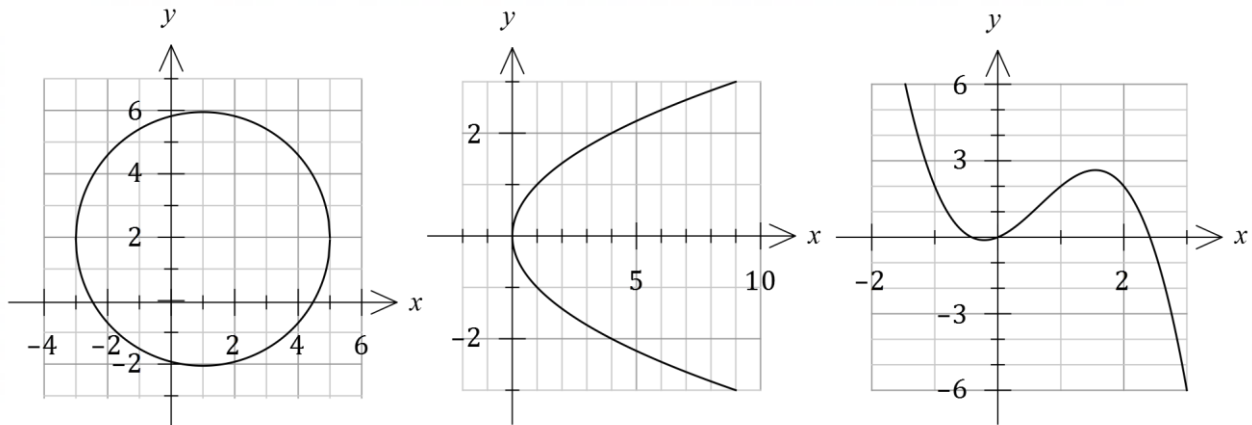
(4 marks)

DO NOT WRITE IN THIS AREA AS IT WILL BE CUT OFF

Question 4

(8 marks)

The graphs of the function $y = f(x)$ and two relations are shown below.



- (a) Explain how the vertical line test can be used to distinguish a function from a relation. (2 marks)
- (b) State the equation of the parabolic relationship. (1 mark)
- (c) Using the graph of the function above determine $f(3)$. (1 mark)
- (d) Using the graph of the function above solve $f(x) = 2$. (1 mark)
- (e) The equation of the above circle is $x^2 + y^2 + ax + by = c$, where a, b and c are constants. Determine the value of each constant. (3 marks)

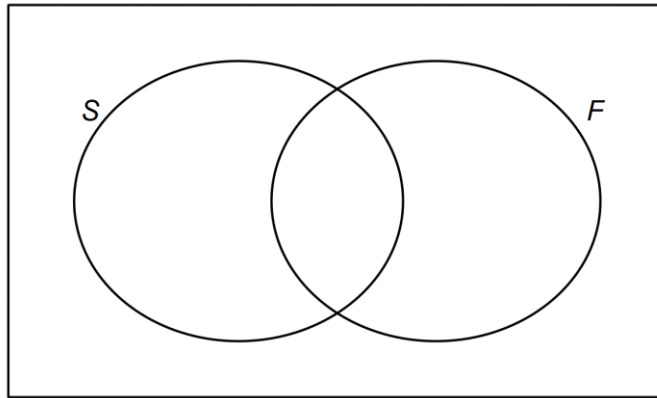
Question 5

(5 marks)

On a certain day, patients report at a general practice to see a doctor.
50% of the patients have a fever.
It is found that the probability of a patient having a sore throat is 0.35,
while the patient having both a fever and a sore throat is 0.1.

- (a) Complete the Probability Venn diagram to represent this situation.

(2 marks)



- (b) What is the probability that a patient selected at random does not have a sore throat?

(1 mark)

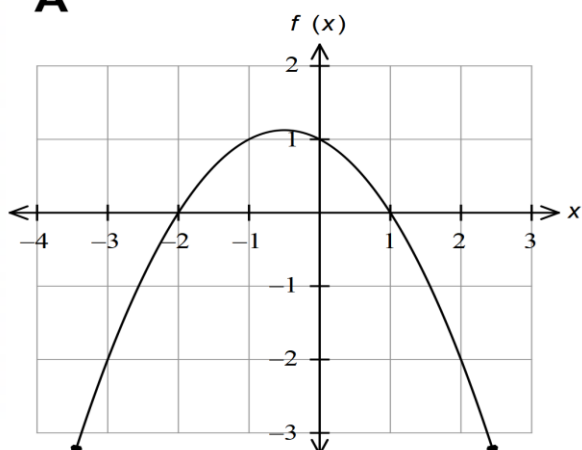
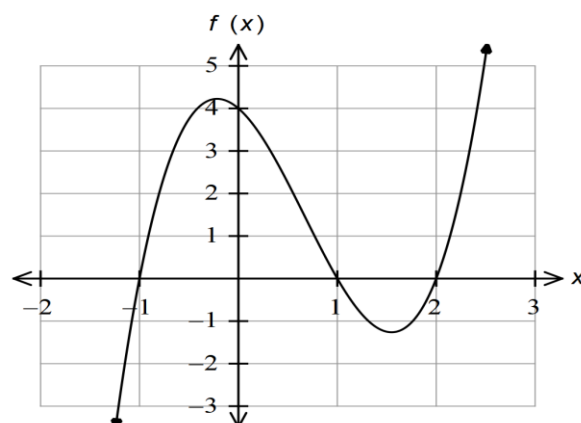
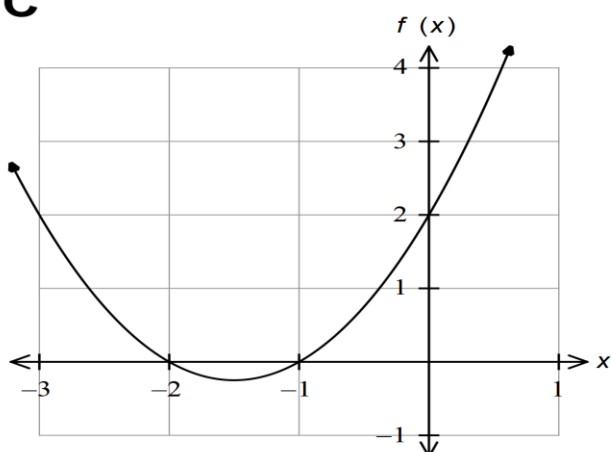
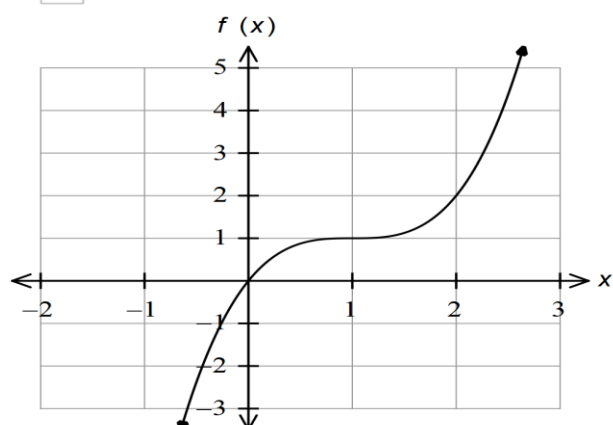
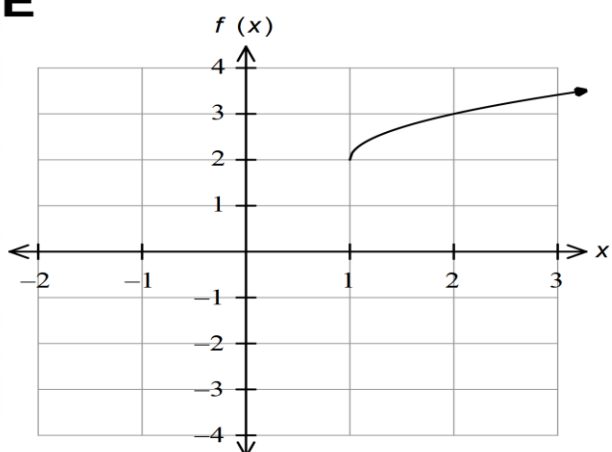
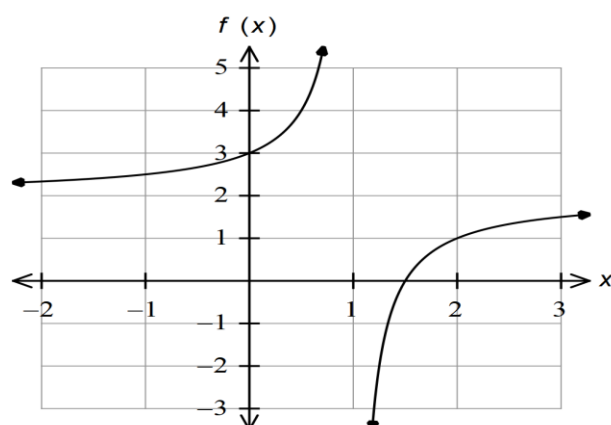
- (c) If a patient selected at random has a sore throat, what is the probability that they have a fever?

(2 marks)

Question 6

(7 marks)

Consider the graphs drawn below.

A**B****C****D****E****F**

The equations of the graphs drawn are given below, in a mixed order.

$f(x) = a(x - 1)(x + 2) \ ;$

$f(x) = \sqrt{(x + b)} + 2 \ ;$

$f(x) = (x - 1)^3 + c$

$f(x) = x^2 + dx + 2 \ ;$

$f(x) = e(x^2 - 1)(x + f) \ ;$

$f(x) = \frac{1}{1 - x} + g$

Determine the values of a, b, c, d, e, f and g , writing your answers in the table below. (7 marks)

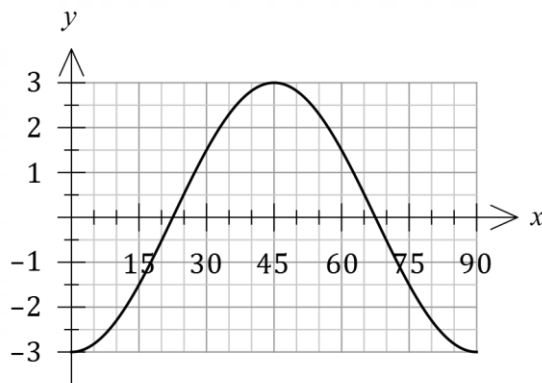
a	b	c	d	e	f	g

DO NOT WRITE IN THIS AREA AS IT WILL BE CUT OFF

Question 7

(14 marks)

- (a) The graph of $y = a \cos(bx)$ is shown. State the value of the constant a and the value of the constant b . (2 marks)



- (b) Point P lies on the unit circle with centre O so that the anticlockwise angle measured from the positive x -axis to the line OP is θ , $0 \leq \theta \leq 2\pi$. Determine the size of θ when P has coordinates $\left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$. (2 marks)

- (c) Solve the equation $1 + \sin(2x - 30^\circ) = 0$ when $0 \leq x \leq 360^\circ$. (3 marks)

- (d) Determine an exact value for $\cos \frac{\pi}{12}$, simplifying fully with a rational denominator. (3 marks)

- (e) Solve $2 \sin \left(x - \frac{\pi}{4} \right) + \sqrt{2} \cos x = 1$ for $0 \leq x \leq 2\pi$. (4 marks)

DO NOT WRITE IN THIS AREA AS IT WILL BE CUT OFF

Supplementary page

Question number: _____

DO NOT WRITE IN THIS AREA AS IT WILL BE CUT OFF

